

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA171/4

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I M. Mohan / 192573011 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: MM

Level 4
60

Annexure A

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1. Solve the Water Jug problem with a 5-gallon jug and a 3-gallon jug (no measuring markers). How can you measure exactly 4 gallons of water using only these two jugs and a pump? List your explicit assumptions and show the sequence of states from start to goal.
2. Given a graph with 8 nodes and a start node S and goal node G, trace Depth-Limited Search (DLS) with a depth limit of 3. Show which nodes are expanded and explain what happens if the goal lies beyond the depth limit.
3. Trace the working of Breadth-First Search and Uniform Cost Search on a weighted graph with at least 7 nodes where edge costs are unequal. Show why BFS may not find the optimal (least-cost) path while UCS does, using your example graph.

1. Water Jug problem

Problem

Measure exactly 4 gallons using a 5-gallon jug, a 3-gallon jug and a pump.

Assumptions

- The pump can fill any jug completely.
- Water can be emptied onto the ground.
- Water can be transferred from one jug to another until one is full or the other is empty.

State representation

State = (5-gallon jug, 3-gallon jug)

Initial state = (0,0)

Goal state = (4,x)

Sequence of status

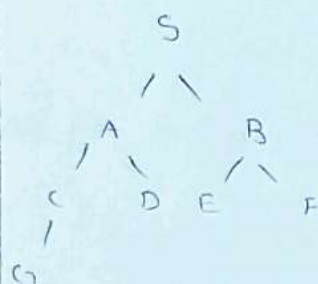
Step	Action	state (5G,3G)
1	Fill 5G jug	(5,0)
2	pour 5G → 3G	(2,3)
3	Empty 3G jug	(2,0)
4	pour remaining 2G → 3G	(0,2)
5	Fill 5G jug	(5,2)
6	pour 5G → 3G until full	(4,3)

Goal achieved:

The 5-gallon jug contains exactly 4 gallons.

2) Depth - Limited search (DLS)

Graph



- Start Node = S
- Goal node = G
- Depth limit = 3

Depth levels

- Level 0 → S
- Level 1 → A, B
- Level 2 → C, D, E, F
- Level 3 → G

DLS Traversal

S → A → C → G

Expanded nodes

S, A, C, G

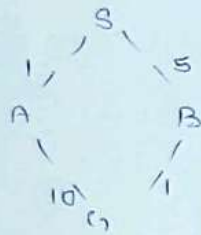
Result:

Goal G is found because it is within the depth limit (Level 3)

If G were at level 4, DLS would stop at level 3 and return cutoff, indicating that the search depth was insufficient even though the may goal exist deeper.

3. Breadth-First search (BFS) vs uniform cost search (UCS)

Weighted graph



Edge costs

- $S \rightarrow A = 1$
- $A \rightarrow G = 10$
- $S \rightarrow B = 5$
- $B \rightarrow G = 1$

Breadth-First search (BFS)

BFS explores level by level

Traversal:

$S \rightarrow A \rightarrow B \rightarrow G$

First path found!

$S \rightarrow A \rightarrow G$

cost = $1 + 10 = 11$

uniform cost search (UCS)

UCS expands the node with the lowest cumulative cost.

Expansion:

- $S(0)$
- $A(1)$
- $B(5)$
- G through $B(6)$

Optimal path:

$S \rightarrow B \rightarrow G$

Total cost = $5 + 1 = 6$

Comparison

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Feature	BFS	UCS
Search strategy	Level-wise	Lowest path cost first
Edge costs	Ignore costs	consider costs
optimal for w.g	No	Yes
Time complexity	$O(b^d)$	$O(b^d(1 + c^*/\epsilon))$
Best use	Equal edge costs	unequal edge costs

Conclusion:

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BFS returns the first shallowest path and may not applicable produce the least-cost solution when edge costs differ. UCS always finds the optimal (least-cost) path by expanding the node with the smallest cumulative cost.

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